

Anupam Bisht

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EDUCATION

- 2021 – Present **PhD Candidate (Expected by Fall, 2024)**
Department of Biomedical Engineering, University of Calgary, Calgary, T2N 1N4, Canada
- Thesis: Devices to enable preclinical mechanistic studies in freely moving rodents.
Supervisor: Prof. Kartikeya Murari
Doctoral Committee: Prof. Zelma Kiss, Prof. Patrick Whelan
- 2019-2021 **Master of Science in Biomedical engineering (Transferred to PhD)**
Department of Biomedical Engineering, University of Calgary, Calgary, Canada
- 2015 - 2019 **Bachelor of Technology in Electrical and Electronics Engineering**
Vellore institute of technology, Vellore, India.

RESEARCH EXPERIENCE

- 2019 Sept - Present **Graduate student at Prof. Kartikeya Murari's Lab, University of Calgary, Calgary, Canada**
- Design and development of electronic devices for neuroscience studies. I am working on a (i) miniaturized wireless electrical stimulation device, (ii) miniaturized wireless optical recording device, both for in-vivo applications, (iii) development of single fiber system for use in in-vivo imaging.
 - Experienced in designing protocols for performing effective pre-clinical experiments.
 - Experienced in microcontroller programming languages (PIC, Arduino) using assembly language.
 - Monte Carlo simulations for investigating optical design specific light propagation through tissue.
 - Experienced in data analysis, inferential statistics, machine learning.
 - Computer interfacing of data acquisition system and hardware.
 - Setting up and synchronization of multiple behavioral equipment (e.g., cameras, stimulus generators) for freely moving rodent experiments (optical recording, electrical stimulation).
 - Building patch cords and custom fiber ferrules for in-vivo imaging of rodents.
 - Animal Handling (Rats, Mouse) during experiments.
 - Intermediate level experience in animal terminal stereotactic animal surgeries for experiments.

- Rat perfusions, cresyl violet staining, immunohistochemistry and confocal brightfield and fluorescence imaging in slide scanner.

2020 Dec – 2019 April

Undergraduate thesis project student in the lab of Prof. Ramasubba Reddy Machireddy, Department of Applied Mechanics, Indian Institute of Technology Madras (IITM), Chennai, India

- Used a Low-cost electroencephalograph (EEG) amplifier built using ADS1299 chip to perform EEG experiments.
- Designed a novel stimulus motivated to overcome neural adaptation of the visual cortex during steady state visually evoked stimulation (SSVEP).
- Brain computer interface-based metrics such as accuracy of target classification (e.g., canonical correlation analysis and Fast Fourier transform based methods), and information transfer rate (ITR) were used to evaluate the stimuli.

2017 - 2018

Undergraduate project in the lab of Prof. Geethanjali Purushothaman School of electrical engineering, Vellore institute of technology, Vellore, India

- Learnt to acquire electroencephalograph (EEG) using 10-20 system of electrode placement.
- Used psychophysics toolbox in MATLAB to generate variety of stimulus for steady state visually evoked stimulation (SSVEP) based brain computer interface (BCI).
- Used Canonical correlation analysis, Fast Fourier transform based methods to evaluate BCI performance.

May-June, 2018

Summer Internship student in the lab of Prof. Aditya Murthy, Center for Neuroscience, Indian Institute of Science (IISc), Bangalore, India

- Learnt about the oculomotor system in the perspective of control theory and analyzed saccade data from experiments.
- Participated in a visuomotor study and learnt to analyze the data from kinarm setup.

June 2017

Summer Internship at Advanced Manufacturing Technology Development Center, Indian Institute of Technology Madras (IITM) Research Park

- Developed an algorithm to segment power signals acquired from grinding machines for a project related to automating grinding machines.

PUBLICATIONS

1. Bisht, Anupam, et al. "Investigating Presence of Motion Artifacts in the Oxygen Saturation Signal during In-Vivo Fiber Photometry." Neural Imaging and Sensing 2022, edited by Qingming Luo et al., SPIE, 2022, p. 8. DOI.org (Crossref), <https://doi.org/10.1117/12.2609877>.

2. Bisht, Anupam, et al. "A New 360° Rotating Type Stimuli for Improved SSVEP Based Brain Computer Interface." Biomedical Signal Processing and Control, vol. 57, Mar. 2020, p. 101778. ScienceDirect, <https://doi.org/10.1016/j.bspc.2019.101778>.

CONFERENCES

2022 Presented a paper at Neural Imaging and Sensing, Photonics West 2022, San Francisco, United States

TEACHING ASSISTANT (at University of Calgary, Canada)

2020, 2021, 2022,2023	Electronic Devices and Materials • Marking exams, debugging hardware circuits (online during the pandemic) in lab activities
2022	Professional Development • Involving students in group discussion for their final project. • Providing periodic project feedbacks and evaluation.
2020, 2021, 2022	Circuits II • Marking exams, debugging hardware circuits (online during the pandemic) in lab activities.
2020	Digital Circuits • Marking exams, evaluating labs.
2021	Biomedical Signals, Systems and Instrumentation I • Helped prepare a lab for the new course.

ONLINE COURSES

2017 Neural Networks and Deep Learning by deeplearning.ai on Coursera. Certificate earned on October 17, 2017.(coursera.org/verify/S4WNGF4GK2CB)

FELLOWSHIPS AND AWARDS (University of Calgary Canada)

2020,2021, 2022	2022 BME Research Award
2022	Robert B. Paugh Memorial Scholarship in Engineering
2020, 2021	Alberta Graduate Excellence Scholarship
2021	Biomedical Engineering Travel Award
2018	Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR) summer fellowship
2015-2018	Merit Scholarship, Vellore Institute of Technology, Vellore, India

PROFESSIONAL SERVICE

- Reviewer for Journal of Biomedical Signal Processing and Control

PROFESSIONAL MEMBERSHIPS

- SPIE

OTHER ACTIVITIES

- Participated in BCI game Jam as a Team organized by the BCI4KIDS group, Alberta Children's Hospital, Calgary, Canada.
- We developed a brain computer interface game using the steady state visually evoked potential paradigm. My role was to develop a python script to perform data acquisition using pyLSL (Lab streaming layer) from a commercial EEG device, and enable real time classification using canonical correlation analysis.

VOLUNTEER AND OUTREACH ACTIVITIES

- Mentor for an incoming international graduate student at the International Student Mentorship Program, University of Calgary, Canada (August-December 2020)
- Presenter for 2 sessions at East Brook Elementary School, Brooks, Alberta, Canada as part of a outreach program of Calgary Optics and Photonics Society. We reached out to about 150 students in total during both the sessions. (October 2019)
- A volunteer at Kulethi primary school Champawat, Uttarakhand, India. I helped to promote tablet assisted teaching in the primary school. I also helped in video making of basic science experiments in Hindi, which are uploaded in YouTube. ([Link](#), June 2016)